

[Update on Clostridium difficile infections].

PMID: 18023937 | Indexed in PubMed

Clostridium difficile is an anaerobic gram positive, spore-forming bacterium which is responsible for 15-25% of antibiotic-associated diarrhea and for more than 95% of pseudomembranous colitis (PMC). This paper will review the main knowledge on *C. difficile*-associated infections and their recent evolution. Since 2003, outbreaks of severe *C. difficile*-associated diarrhea (CDAD) have been increasingly reported in Canada and the United States. This trend is assumed to be associated with the rapid emergence and spread of a specific clone of *C. difficile* belonging to PCR-ribotype 027 or North American Pulsotype 1, pulsotype (NAP1). This clone is characterized by the overproduction of toxins A and B and is positive for a third toxin named binary toxin. This clone has spread in UK, in Belgium, in the Netherlands, and, more recently, in France where it has been responsible for large outbreaks mainly in northern France. A systematic reporting of *C. difficile* incidence by health facilities should enable a better assessment of this pathology in France.

The microbial metabolite urolithin A reduces Clostridioides difficile toxin expression and toxin-induced epithelial damage.

PMID: 38193707 | Indexed in PubMed

Clostridioides difficile is a Gram-positive, anaerobic, spore-forming bacterium responsible for antibiotic-associated pseudomembranous colitis. *Clostridioides difficile* infection (CDI) symptoms can range from diarrhea to life-threatening colon damage. Toxins produced by *C. difficile* (TcdA and TcdB) cause intestinal epithelial injury and lead to severe gut barrier dysfunction, stem cell damage, and impaired regeneration of the gut epithelium. Current treatment options for intestinal repair are limited. In this study, we demonstrate that treatment with the microbial metabolite urolithin A (UroA) attenuates CDI-induced adverse effects on the colon epithelium in a preclinical model of CDI-induced colitis. Moreover, our analysis suggests that UroA treatment protects against *C. difficile*-induced inflammation, disruption of gut barrier integrity, and intestinal tight junction proteins in the colon of CDI mice. Importantly, UroA treatment significantly reduced the expression and release of toxins from *C. difficile* without inducing bacterial cell death. These results indicate the direct regulatory effects of UroA on bacterial gene regulation. Overall, our findings reveal a novel aspect of UroA activity, as it appears to act at both the bacterial and host levels to protect against CDI-induced colitis pathogenesis. This research sheds light on a promising avenue for the development of novel treatments for *C. difficile* infection. IMPORTANCE Therapy for *Clostridioides difficile* infections includes the use of antibiotics, immunosuppressors, and fecal microbiota transplantation. However, these treatments have several drawbacks, including the loss of colonization resistance, the promotion of autoimmune disorders, and the potential for unknown pathogens in donor samples. To date, the potential benefits of microbial metabolites in CDI-induced colitis have not been fully investigated. Here, we report for the first time that the microbial metabolite urolithin A has the potential to block toxin production from *C. difficile* and enhance gut barrier function to mitigate CDI-induced colitis.

The microbial metabolite Urolithin A reduces C. difficile toxin expression and repairs toxin-induced epithelial damage.

PMID: 37546803 | Indexed in PubMed

Clostridioides difficile is a gram-positive, anaerobic, spore-forming bacterium that is responsible for antibiotic-associated pseudomembranous colitis. *Clostridioides difficile* infection (CDI) symptoms can range from diarrhea to life-threatening colon damage. Toxins produced by *C. difficile* (TcdA and TcdB) cause intestinal epithelial injury and lead to severe gut barrier dysfunction, stem cell damage, and impaired regeneration of the gut epithelium. Current treatment options for intestinal repair are limited. In this study, we demonstrate that treatment with the microbial metabolite urolithin A (UroA) attenuates CDI-induced adverse effects on the colon epithelium in a preclinical model of CDI-induced colitis. Moreover, our analysis suggests that UroA treatment protects against *C. difficile*-induced inflammation, disruption of gut barrier integrity, and intestinal tight junction proteins in the colon of CDI mice. Importantly, UroA treatment significantly reduced the expression and release of toxins from *C. difficile*, without inducing bacterial cell death. These results indicate the direct regulatory effects of UroA on bacterial gene regulation. Overall, our findings reveal a novel aspect of UroA activities, as it appears to act at both the bacterial and host levels to protect against CDI-induced colitis pathogenesis. This research sheds light on a promising avenue for the development of novel treatments for *C. difficile* infection.

Emergence of a non-sporulating secondary phenotype in *Clostridium* (*Clostridioides*) *difficile* ribotype 078 isolated from humans and animals.

PMID: 31548637 | Indexed in PubMed

Clostridium (*Clostridioides*) *difficile* is a Gram positive, spore forming anaerobic bacterium that is a leading cause of antibiotic associated diarrhoea in the developed world. *C. difficile* is a genetically diverse species that can be divided into 8 phylogenetically distinct clades with clade 5 found to be genetically distant from all others. Isolates with the PCR ribotype 078 belong to clade 5, and are often associated with *C. difficile* infection in both humans and animals. Colonisation of animals and humans by ribotype 078 raises questions about possible zoonotic transmission, and also the diversity of reservoirs for ribotype 078 strains within the environment. One of the key factors which enables *C. difficile* to be a successful, highly transmissible pathogen is its ability to produce oxygen resistant spores capable of surviving harsh conditions. Here we describe the existence of a non-sporulating variant of *C. difficile* ribotype 078 harbouring mutations leading to premature stop codons within the master regulator, Spo0A. As sporulation is imperative to the successful transmission of *C. difficile* this study was undertaken to investigate phenotypic characteristics of this asporogenous phenotype with regards to growth rate, antibiotic susceptibility, toxin production and biofilm formation.

Triggering germination represents a novel strategy to enhance killing of *Clostridium difficile* spores.

PMID: 20808871 | Indexed in PubMed

Clostridium difficile is an anaerobic, spore-forming bacterium that is the most common cause of healthcare-associated diarrhea in developed countries. Control of *C. difficile* is challenging because the spores are resistant to killing by alcohol-based hand hygiene products, antimicrobial soaps, and most disinfectants. Although initiation of germination has been shown to increase susceptibility of spores of

other bacterial species to radiation and heat, it was not known if triggering of germination could be a useful strategy to increase susceptibility of *C. difficile* spores to radiation or other stressors. Here, we demonstrated that exposure of dormant *C. difficile* spores to a germination solution containing amino acids, minerals, and taurocholic acid resulted in initiation of germination in room air. Germination of spores in room air resulted in significantly enhanced killing by ultraviolet-C (UV-C) radiation and heat. On surfaces in hospital rooms, application of germination solution resulted in enhanced eradication of spores by UV-C administered by an automated room decontamination device. Initiation of germination under anaerobic, but not aerobic, conditions resulted in increased susceptibility to killing by ethanol, suggesting that exposure to oxygen might prevent spores from progressing fully to outgrowth. Stimulation of germination also resulted in reduced survival of spores on surfaces in room air, possibly due to increased susceptibility to stressors such as oxygen and desiccation. Taken together, these data demonstrate that stimulation of germination could represent a novel method to enhance killing of spores by UV-C, and suggest the possible application of this strategy as a means to enhance killing by other agents.

Flagellin is essential for initial attachment to mucosal surfaces by *Clostridioides difficile*.

PMID: 37292962 | Indexed in PubMed

Mucins are glycoproteins which can be found in host cell membranes and as a gelatinous surface formed from secreted mucins. Mucosal surfaces in mammals form a barrier to invasive microbes, particularly bacteria, but are a point of attachment for others. *Clostridioides difficile* is an anaerobic bacterium which colonizes the mammalian GI tract and is a common cause of acute GI inflammation leading to a variety of negative outcomes. Although *C. difficile* toxicity stems from secreted toxins, colonization is a prerequisite for *C. difficile* disease. While *C. difficile* is known to associate with the mucus layer and underlying epithelium, the mechanisms underlying these interactions that facilitate colonization are less well-understood. To understand the molecular mechanisms by which *C. difficile* interacts with mucins, we used ex vivo mucosal surfaces to test the ability of *C. difficile* to bind to mucins from different mammalian tissues. We found significant differences in *C. difficile* adhesion based upon the source of mucins, with highest levels of binding observed to mucins purified from the human colonic adenocarcinoma line LS174T and lowest levels of binding to porcine gastric mucin. We also observed that defects in adhesion by mutants deficient in flagella, but not type IV pili. These results imply that interactions between host mucins and *C. difficile* flagella facilitate the initial host attachment of *C. difficile* to host cells and secreted mucus.

Environmentally Acquired *Bacillus* and Their Role in *C. difficile* Colonization Resistance.

PMID: 35625667 | Indexed in PubMed

Clostridioides difficile is an environmentally acquired, anaerobic, spore-forming bacterium which ordinarily causes disease following antibiotic-mediated dysbiosis of the intestinal microbiota. Although much is understood regarding the life cycle of *C. difficile*, the fate of *C. difficile* spores upon ingestion remains unclear, and the underlying factors that predispose an individual to colonization and subsequent development of *C. difficile* infection (CDI) are not fully understood. Here, we show that *Bacillus*, a ubiquitous and environmentally acquired, spore-forming bacterium is associated with colonization resistance to *C. difficile*. Using animal models, we first provide evidence that animals housed under conditions that mimic reduced environmental exposure have an increased susceptibility to CDI,

correlating with a loss in *Bacillus*. Lipopeptide micelles (~10 nm) produced by some *Bacilli* isolated from the gastro-intestinal (GI)-tract and shown to have potent inhibitory activity to *C. difficile* have recently been reported. We show here that these micelles, that we refer to as heterogenous lipopeptide lytic micelles (HELMs), act synergistically with components present in the small intestine to augment inhibitory activity against *C. difficile*. Finally, we show that provision of HELM-producing *Bacillus* to microbiota-depleted animals suppresses *C. difficile* colonization thereby demonstrating the significant role played by *Bacillus* in colonization resistance. In the wider context, our study further demonstrates the importance of environmental microbes on susceptibility to pathogen colonization.

Zotiraciclib (TG02) for newly diagnosed glioblastoma in the elderly or for recurrent glioblastoma: The EORTC 1608 STEAM trial.

PMID: 38159337 | Indexed in PubMed

Zotiraciclib (TG02) is an oral multi-cyclin dependent kinase (CDK) inhibitor thought to inhibit tumor growth via CDK-9-dependent depletion of survival proteins such as c-MYC and MCL-1 which are frequently overexpressed in glioblastoma. EORTC 1608 (NCT03224104) (STEAM) had a three parallel group (A,B,C) phase Ib, open-label, non-randomized, multicenter design in IDH wild-type newly diagnosed glioblastoma or anaplastic astrocytoma. Groups A and B explored the maximum tolerated dose (MTD) of TG02 in elderly patients, in combination with hypofractionated radiotherapy alone (group A) or temozolomide alone (group B), according to O6-methylguanine DNA methyltransferase promoter methylation status determined centrally. Group C explored single agent activity of TG02 at first relapse after temozolomide chemoradiotherapy with a primary endpoint of progression-free survival at 6 months (PFS-6). Tumor expression of CDK-9, c-MYC and MCL-1 was determined by immunohistochemistry. The MTD was 150 mg twice weekly in combination with radiotherapy alone (group A) or temozolomide alone (group B). Two dose-limiting toxicities were observed at 150 mg: one in group A (grade 3 seizure), one in group B (multiple grade 1 events). Main toxicities included neutropenia, gastrointestinal disorders and hepatotoxicity. PFS-6 in group C was 6.7%. CDK-9, c-MYC and MCL-1 were confirmed to be expressed and their expression was moderately cross-correlated. High protein levels of MCL-1 were associated with inferior survival. TG02 exhibits overlapping toxicity with alkylating agents and low single agent clinical activity in recurrent glioblastoma. The role of CDK-9 and its down-stream effectors as prognostic factors and therapeutic targets in glioblastoma warrants further study.

Focus on current and emerging treatment options for glioma: A comprehensive review.

PMID: 38689623 | Indexed in PubMed

This comprehensive review delves into the current updates and challenges associated with the management of low-grade gliomas (LGG), the predominant primary tumors in the central nervous system. With a general incidence rate of 5.81 per 100000, gliomas pose a significant global concern, necessitating advancements in treatment techniques to reduce mortality and morbidity. This review places a particular focus on immunotherapies, discussing promising agents such as Zotiraciclib and Lerapolturev. Zotiraciclib, a CDK9 inhibitor, has demonstrated efficacy in glioblastoma treatment in preclinical and clinical studies, showing its potential as a therapeutic breakthrough. Lerapolturev, a viral immunotherapy, induces inflammation in glioblastoma and displays positive outcomes in both adult and

pediatric patients. Exploration of immunotherapy extends to Pembrolizumab, Nivolumab, and Entrectinib, revealing the challenges and variabilities in patient responses. Despite promising preclinical data, the monoclonal antibody Depatuxizumab has proven ineffective in glioblastoma treatment, emphasizing the critical need to understand resistance mechanisms. The review also covers the success of radiation therapy in pediatric LGG, with evolving techniques, such as proton therapy, showing potential improvements in patient quality of life. Surgical treatment is discussed in the context of achieving a balance between preserving the patient's quality of life and attaining gross total resection, with the extent of surgical resection significantly influencing the survival outcomes. In addition to advancements in cancer vaccine development, this review highlights the evolving landscape of LGG treatment, emphasizing a shift toward personalized and targeted therapies. Ongoing research is essential for refining strategies and enhancing outcomes in the management of LGG.

Inactivation of Tumor Suppressor CYLD Inhibits Fibroblast Reprogramming to Pluripotency.

PMID: 37894364 | Indexed in PubMed

CYLD is a tumor suppressor gene coding for a deubiquitinating enzyme that has a critical regulatory function in a variety of signaling pathways and biological processes involved in cancer development and progression, many of which are also key modulators of somatic cell reprogramming. Nevertheless, the potential role of CYLD in this process has not been studied. With the dual aim of investigating the involvement of CYLD in reprogramming and developing a better understanding of the intricate regulatory system governing this process, we reprogrammed control (CYLDWT/WT) and CYLD DUB-deficient (CYLD Δ 9/ Δ 9) mouse embryonic fibroblasts (MEFs) into induced pluripotent stem cells (iPSCs) through ectopic overexpression of the Yamanaka factors (Oct3/4, Sox2, Klf4, c-myc). CYLD DUB deficiency led to significantly reduced reprogramming efficiency and slower early reprogramming kinetics. The introduction of WT CYLD to CYLD Δ 9/ Δ 9 MEFs rescued the phenotype. Nevertheless, CYLD DUB-deficient cells were capable of establishing induced pluripotent colonies with full spontaneous differentiation potential of the three germ layers. Whole proteome analysis (Data are available via ProteomeXchange with identifier PXD044220) revealed that the mesenchymal-to-epithelial transition (MET) during the early reprogramming stages was disrupted in CYLD Δ 9/ Δ 9 MEFs. Interestingly, differentially enriched pathways revealed that the primary processes affected by CYLD DUB deficiency were associated with the organization of the extracellular matrix and several metabolic pathways. Our findings not only establish for the first time CYLD's significance as a regulatory component of early reprogramming but also highlight its role as an extracellular matrix regulator, which has profound implications in cancer research.